Spanning Tree

A spanning tree is a subset of Graph G, which has all the vertices covered with minimum possible number of edges. Hence, a spanning tree does not have cycles and it cannot be disconnected..

Properties of Spanning Tree

- A connected graph G can have more than one spanning tree.
- Spanning Tree has n-1 edges, where n is the number of Node (Vertices).
- All possible spanning trees of graph G, have the same number of edges and vertices.
- The spanning tree does not have any cycle (loops).
- Removing one edge from the spanning tree will make the graph disconnected, i.e. the spanning tree is minimally connected.
- Adding one edge to the spanning tree will create a circuit or loop, i.e. the spanning tree is maximally acyclic.

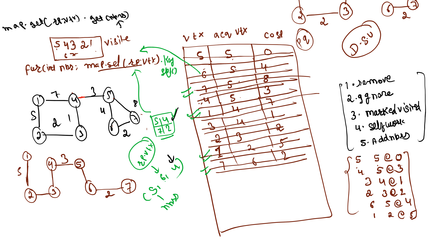
Minimum Spanning Tree (MST)
In a weighted graph, a minimum spanning tree is a spanning tree that has minimum weight than all other spanning trees of the same graph.

Kruskal's Algorithm

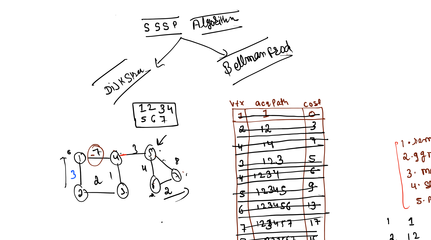
Prim's Algorithm
Both are greedy algorithms.

Consider the following graph:

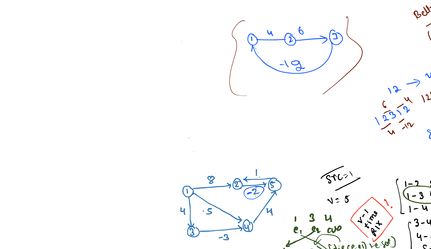
Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm?
(A) (b,e),(e,f),(a,c),(b,c),(f,g),(c,d)
(B) (b,e),(e,f),(a,c),(f,g),(b,c),(c,d)
(C) (b,e),(a,c),(e,f),(b,c),(f,g),(c,d)
(D) (b,e),(e,f),(b,c),(a,c),(f,g),(c,d)

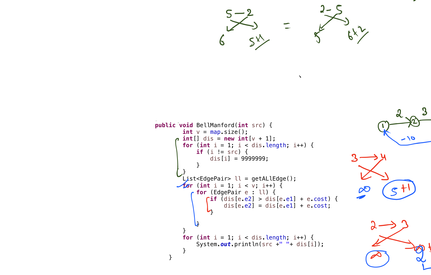


vtx, acq, vtx, cost




```
public void Dijkstra_Algo(int src) {
    PriorityQueue<DijkstraPair> pq = new PriorityQueue<>((a,b)->a.cost-b.cost);
    HashSet<Integer> visited = new HashSet<>();
    pq.add(new DijkstraPair(src, "" + src, 0));
    while (!pq.isEmpty()) {
        // 1. remove
        DijkstraPair rp=pq.poll();
        // 2. Ignore if Already visited
        if(visited.contains(rp.vtx))
            continue;
        }
        // 3. Marked visited
        visited.add(rp.vtx);
        // 4. Self work
        System.out.println(rp);
        // 5. Add nbrs
        for (int nbrs : map.get(rp.vtx).keySet()) {
            if (!visited.contains(nbrs)) {
                int cost = map.get(rp.vtx).get(nbrs);
                pq.add(new DijkstraPair(nbrs, rp.acq_path+nbrs, rp.cost+cost));
            }
        }
    }
}
```






```
public void BellManford(int src) {
    int v = map.size();
    int[] dis = new int[v + 1];
    for (int i = 1; i < dis.length; i++) {
        if (i != src) {
            dis[i] = 9999999;
        }
    }
    List<EdgePair> ll = getALLEdge();
    for (int i = 1; i < v; i++) {
        for (EdgePair e : ll) {
            if (dis[e.e2] > dis[e.e1] + e.cost) {
                dis[e.e2] = dis[e.e1] + e.cost;
            }
        }
    }
    for (int i = 1; i < dis.length; i++) {
        System.out.println(src + "->" + dis[i]);
    }
}
```